

EVA Humanoid Robot
3-Year Consolidated Project Report

SRM Institute of Science and Technology
College of Engineering & Technology
2024 – 2026

(A Multidisciplinary Robotics Project integrating Mechanical Engineering,
Electronics & Embedded Systems, Artificial Intelligence, NLP/LLM, and UI/UX)

CHAPTER 1 – INTRODUCTION

EVA (Enhanced Visitor Assistant) is a multidisciplinary humanoid robotics project developed over three years with the objective of creating an intelligent, expressive, and interactive humanoid capable of serving in public spaces such as universities, receptions, exhibitions, and labs. The project combines mechanical design, advanced fabrication, autonomous navigation, NLP-driven communication, bionic facial expressions, and a fully custom UI.

Over the span of three years, EVA has evolved through three major versions:

- EVA 1.0 – Foundation: drive system, skeletal frame, SLAM prototype, basic arms.
- EVA 2.0 – Refinement: BLDC arms, CAN PCBs, NAV2, facial expression system.
- EVA 3.0 – Completion: full shell design, Teensy-controlled arms, bionic head, multilingual LLM-NLP, new AMOLED UI.

This report consolidates all three development phases into a comprehensive 25-page technical document.

CHAPTER 2 – SYSTEM EVOLUTION OVER THREE YEARS

2.1 EVA 1.0 (Year 1 – 2023)

EVA 1.0 established the fundamental architecture of the robot. The initial version focused primarily on:

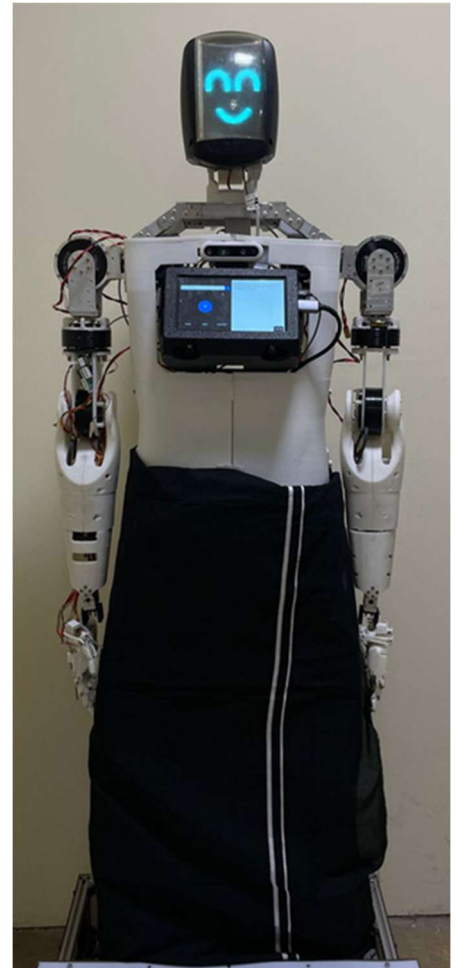
- Omni-wheel drive system (planetary DC motors with encoders)
- Basic ROS2 navigation stack
- Jetson Xavier-based AI
- Simple UI through a small display
- Lightweight PLA arm prototypes with servo actuation



2.2 EVA 2.0 (Year 2 – 2024)

EVA 2.0 transitioned the platform towards a more advanced humanoid with realistic motion and autonomous behavior:

- High-torque back-drivable BLDC motors (AK70-10, AK60-6)
- CAN-based custom arm PCB and distributed control
- NAV2 for autonomous indoor navigation
- PD-controlled arm gestures (hello, namaste, handshake)
- Early expressive head using LED-based facial animations



2.3 EVA 3.0 (Year 3 – 2025)

EVA 3.0 marks the transition into a deployable humanoid system:

- Full external shell and body aesthetics
- Teensy-driven real-time BLDC arm control replacing Raspberry Pi CAN-HAT
- 13.3-inch AMOLED touchscreen with custom UI
- Bionic head with actuated eyes, lips, cheeks for human-like expressions
- Beamforming microphone array for robust multilingual voice recognition
- Full LLM-powered NLP with contextual memory and SRM campus information

CHAPTER 3 – MECHANICAL DESIGN AND FABRICATION

3.1 Drive System

The drive is a four-wheel omnidirectional platform designed with a custom cyclic wheel configuration. Compared to mecanum wheels, this design decreases roller wear and increases long-term reliability.

Key Features:

- High torque DC planetary motors (Mega Torque 750W)
- Encoders for velocity feedback
- Cytron MDDS30A motor drivers
- IMU-based odometry correction

3.2 Arm Mechanics (6-DOF)

The arms feature a 6-DOF structure actuated with backdrivable BLDC motors offering safe and smooth human-like movements. The redesigned forearm integrates lightweight structural support, improved torque distribution, and custom CAN node integration.

3.3 URDF Simulation Model

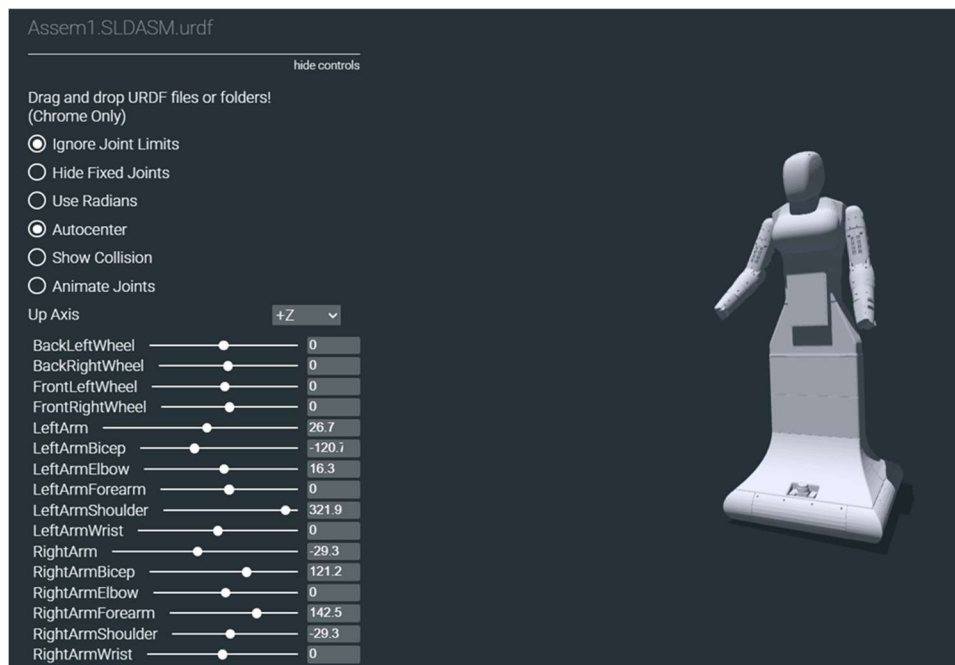
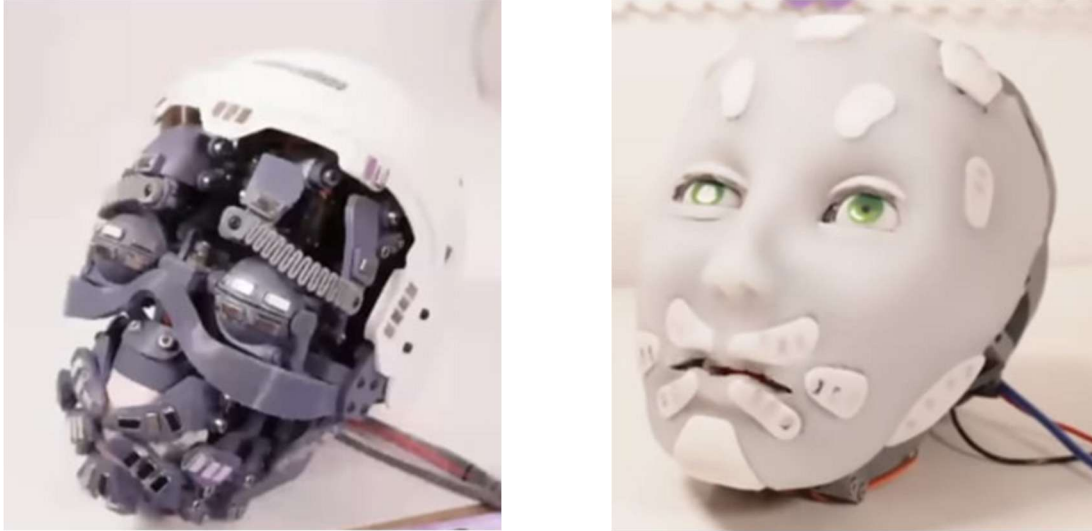


Figure: URDF visualization of EVA showing joint-level simulation and motion validation.

3.4 Bionic Head System

The most advanced subsystem in EVA 3.0 is the bionic head. It replaces the earlier LED-display face with a fully animatronic mechanism. With multiple micro-servos controlling eyelids, eyeballs, lips, and cheeks, the system achieves naturalistic expressions.

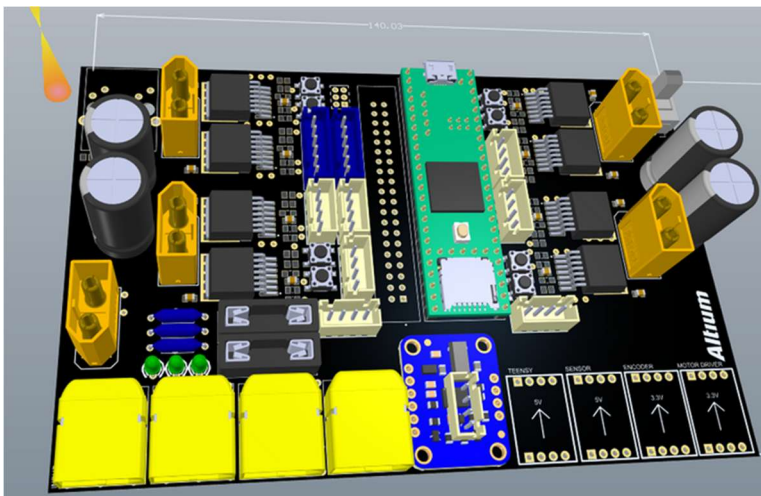


Figures: EVA 3.0 bionic face outer skin and internal mechanical structure.

CHAPTER 4 – ELECTRONICS & EMBEDDED SYSTEMS

4.1 Teensy-Based BLDC Controller

The Year 3 redesign moves all arm actuation to a Teensy 4.1 microcontroller for low-latency FOC communication via CAN. A custom PCB integrates power regulation, encoder interfaces, and dual BLDC drivers.



4.2 Sensor Suite

EVA integrates the following sensors:

- Beamforming microphone array
- IMU (BNO055)
- Encoders for arm joints and drive wheels
- LIDAR for SLAM and mapping

CHAPTER 5 – SOFTWARE ARCHITECTURE

5.1 ROS2 Framework

EVA's software core is built on ROS 2 Humble, using a modular node-based architecture. Subsystems communicate through dedicated topics for navigation, arm control, NLP, and UI.

5.2 Autonomous Navigation

NAV2 provides mapping, localization, path planning, and obstacle avoidance. Custom wheel kinematics enable omni-directional movement. SLAM Toolbox is used for map construction.

5.3 LLM-NLP System

The NLP pipeline in EVA 3.0 includes:

- Beamforming microphone preprocessing
- Whisper-based speech recognition
- LLaMA 3-based multilingual LLM for dialogue
- Emotion classification
- Text-to-speech for natural response generation

CHAPTER 6 – USER INTERFACE & EXPERIENCE

6.1 13.3-inch AMOLED Interface

A high-brightness AMOLED touchscreen powers EVA's new UI. The interface includes menus for campus navigation, faculty search, FAQs, and voice-based interaction.

6.2 UI Redesign Philosophy

The new UI follows a minimalistic, human-friendly layout with large icons, haptic feedback, animated transitions, and accessibility features.

CHAPTER 7 – PROJECT TIMELINE (2023–2025)

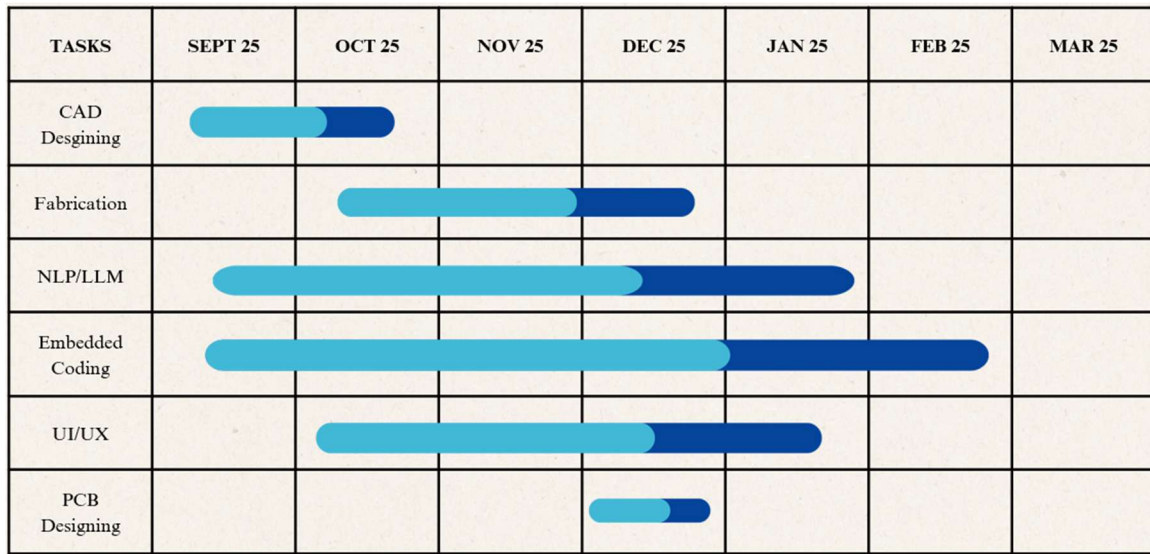


Figure: Consolidated task timeline showing design, fabrication, coding, NLP, and PCB phases.

CHAPTER 8 – RESULTS & TESTING

8.1 Mechanical Results

- Smooth arm motion with IK + PD control
- Improved torque distribution
- Reduced vibration in omni-drive

8.2 NLP Results

- Multilingual communication functioning
- Accurate speech pickup using beamforming microphone
- Faster inference on Jetson AGX Orin

CHAPTER 9 – CONCLUSION

Over three years of iterative development, EVA evolved from a simple prototype into a nearly deployable humanoid robot. With a fully actuated bionic head, 6-DOF BLDC arms, autonomous navigation, multilingual NLP, and a modern user interface, EVA stands as one of the most advanced student-built humanoids at SRM IST.